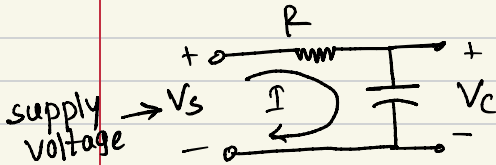


Signal Generators

→ Basic R-C ckt:



if $V_s > V_c \rightarrow$ capa store charge
if $V_s < V_c \rightarrow$ capa discharge

Suppose at initial time t_1
the capacitor has a voltage of
 $V_c(t_1) = V_{\text{initial}}$

for capacitor $\rightarrow \boxed{Q = CV}$

$$\hookrightarrow \text{Current, } I = \frac{dQ}{dt} = \frac{d(CV)}{dt} = C \frac{dV_c}{dt} \quad \text{--- (i)}$$

$$\text{from ckt} \rightarrow I = \frac{V_s - V_c}{R} \quad \text{--- (ii)}$$

equating (i) & (ii)

$$C \frac{dV_c}{dt} = \frac{V_s - V_c}{R} \quad \text{--- (iii)}$$

We can determine the voltage across capacitor, $V_c(t)$
for the time $t_1 < t < \infty$, by solving the Differential
eqn (iii) -

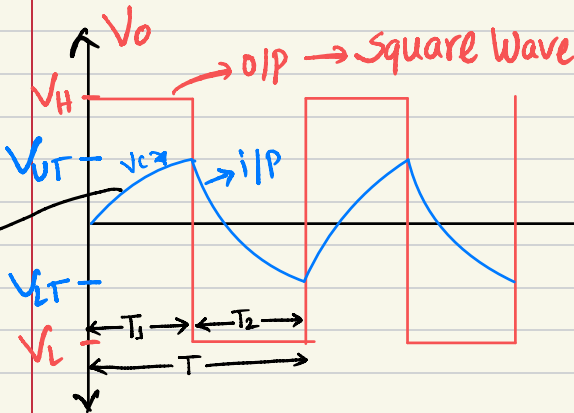
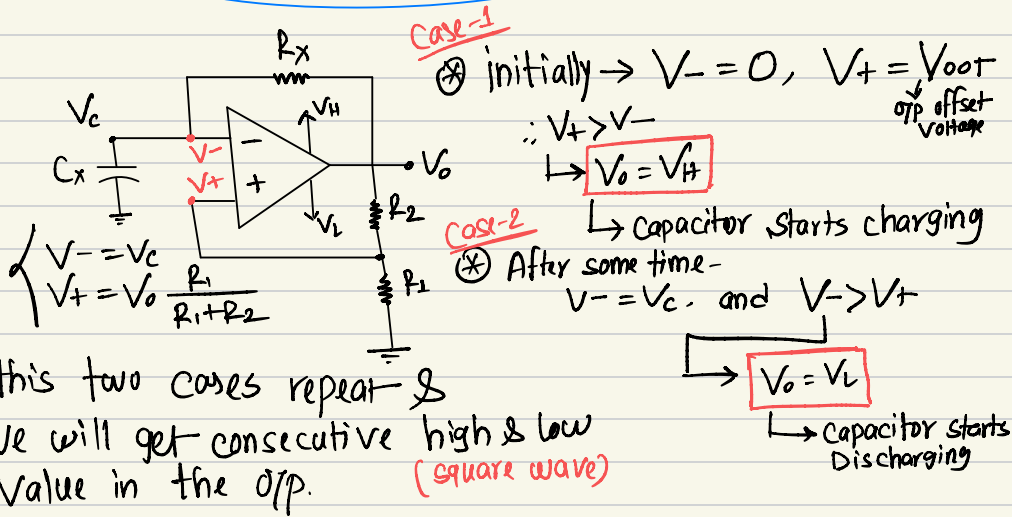
$$\begin{aligned} C \frac{dV_c}{dt} &= \frac{V_s - V_c}{R} \\ \Rightarrow \int_{V_c(t_1)}^{V_c(t)} \frac{dV_c}{V_s - V_c} &= \int_{t_1}^t \frac{dt}{RC} \Rightarrow V_c(t) = V_s + (V_c(t_1) - V_s) \exp\left(-\frac{t-t_1}{RC}\right) \\ \text{at } t \rightarrow \infty \quad V_c(\infty) &= V_s \rightarrow V_{\text{final}} \end{aligned}$$

Time constant, $\tau = RC$

$$\boxed{V_c(t) = V_{\text{final}} + (V_{\text{initial}} - V_{\text{final}}) \exp\left(-\frac{t-t_1}{\tau}\right)}$$

Schmitt Oscillator / Astable Multivibrator:

→ Square Wave generator.



$$V_{UT} = \frac{R_1}{R_1 + R_2} V_H$$

$$V_{LT} = \frac{R_1}{R_1 + R_2} V_L$$

period, $T = T_1 + T_2$, → Duty cycle, $D = \frac{T_1}{T_1 + T_2} \times 100\%$

→ Time const, $\tau = R_x C_x$

→ $V_c(t) = V_{final} + (V_{initial} - V_{final}) \exp\left(-\frac{t-t_1}{\tau}\right)$, $V_{final} = V_o$

$\therefore t - t_1 = \tau \ln \left| \frac{V_o - V_c(t_1)}{V_o - V_c(t_2)} \right|$

Case-1 : $0 < t < T_1$

$$V_{\text{final}} = V_0 = V_H, \quad V_{\text{initial}} = V_{LT}$$

$$V_C(t) = V_H + (V_{LT} - V_H) \exp\left(-\frac{t-0}{R_X C_X}\right)$$

$$\text{At } t = T_1, V_C(t) = V_{UT}$$

$$\therefore V_C(T_1) = V_{UT} = V_H + (V_{LT} - V_H) \exp\left(-\frac{T_1}{R_X C_X}\right)$$

$$\therefore T_1 = R_X C_X \ln \left| \frac{V_H - V_{LT}}{V_H - V_{UT}} \right|$$

Case-2 : $T_1 < t < T$

$$V_{\text{final}} = V_0 = V_L, \quad V_{\text{initial}} = V_{UT}$$

$$\therefore V_C(t) = V_L + (V_{UT} - V_L) \exp\left(-\frac{t-T_1}{R_X C_X}\right)$$

$$\text{At } t = T \rightarrow V_C(t) = V_{LT}$$

$$\therefore V_C(t) = V_{LT} = V_L + (V_{UT} - V_L) \exp\left(-\frac{T-T_1}{R_X C_X}\right)$$

$$V_{LT} = V_L + (V_{UT} - V_L) \exp\left(-\frac{T_2}{R_X C_X}\right)$$

$$\therefore T_2 = R_X C_X \ln \left| \frac{V_L - V_{UT}}{V_L - V_{LT}} \right|$$

$$T = T_1 + T_2$$

Practice Problem:

For the Schmitt trigger oscillator, the saturation output voltages are $+10\text{V}$ and -5V . $R_1 = R_2 = 20\text{K}\Omega$, $R_x = 50\text{K}\Omega$ and $C_x = 0.01\mu\text{F}$.

Determine the frequency and duty cycle.

⇒ Solⁿ:

$$V_H = 10\text{V}, V_L = -5\text{V}$$

$$V_{UT} = \frac{R_1}{R_1 + R_2} V_H = \frac{20}{20 + 20} \times 10 = 5\text{V}$$

$$V_{LT} = \frac{R_1}{R_1 + R_2} V_L = \frac{20}{20 + 20} (-5) = -2.5\text{V}$$

$$\tau = R_x C_x = 50 \times 10^3 \times 0.01 \times 10^{-6} = 0.5 \times 10^{-3} \text{ s} \\ = 0.5 \text{ ms}$$

$$\rightarrow \text{now, } T_1 = \tau \ln \left| \frac{V_H - V_{LT}}{V_H - V_{UT}} \right| \\ = 0.5 \ln \left| \frac{10 - (-2.5)}{10 - 5} \right| = 0.458 \text{ ms}$$

$$\text{and, } T_2 = \tau \ln \left| \frac{V_L - V_{UT}}{V_L - V_{LT}} \right| = 0.5 \ln \left| \frac{-5 - 5}{-5 + 2.5} \right| \\ = 0.693 \text{ ms}$$

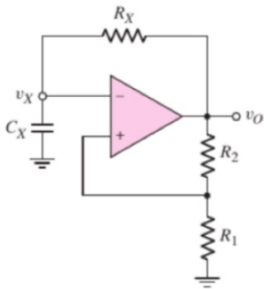
$$\therefore \text{Period, } T = T_1 + T_2 = 1.15 \text{ ms}$$

$$\therefore \text{frequency, } f = \frac{1}{T} = 0.868 \text{ kHz} = 868 \text{ Hz}$$

$$\therefore \text{Duty cycle, } D = \frac{T_1}{T} \times 100\% = 39.82\%$$

(Ans.)

Practice Problem-2



Here, $V_H = 10\text{V}$ and $V_L = -10\text{V}$
 Considering $R_1 = R_2$, design the circuit so that it can generate a square wave with 1kHz freq and 50% duty cycle.

⇒ Soln: $R_1 = R_2$, $V_H = 10\text{V}$, $V_L = -10\text{V}$

$$V_{UT} = \frac{R_1}{R_1 + R_2} V_H = \frac{1}{2} \times 10 = 5\text{V}, \quad V_{LT} = \frac{R_1}{R_1 + R_2} \times V_L = -5\text{V}$$

$$f = 1\text{kHz} = 1000\text{Hz}, \quad D = 50\%$$

$$\therefore T = \frac{1}{f} = 10^{-3}\text{s} = 1\text{ms}$$

$$D = \frac{T_1}{T_1 + T_2} \times 100\%$$

$$\Rightarrow 50\% = \frac{T_1}{T} \times 100\%$$

$$\Rightarrow 0.5 = \frac{T_1}{1\text{ms}} \quad \therefore T_1 = 0.5\text{ms}$$

$$T_1 = R_X C_X \ln \left| \frac{V_H - V_{LT}}{V_H - V_{UT}} \right|$$

$$\Rightarrow 0.5 \times 10^{-3} = R_X C_X \ln \left| \frac{10 - (-5)}{10 - 5} \right|$$

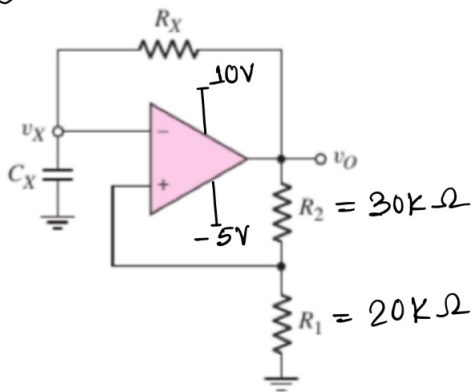
$$\Rightarrow R_X C_X = \frac{0.5 \times 10^{-3}}{\ln(3)} = 4.5 \times 10^{-4}\text{s}$$

$$\text{let } R_X = 10\text{k}\Omega$$

$$\therefore C_X = \frac{4.5 \times 10^{-4}}{R_X} = \frac{4.5 \times 10^{-4}}{10^4} = 4.5 \times 10^{-8}\text{F} = 45 \times 10^{-9}\text{F}$$

$$\boxed{C_X = 45\text{nF}}$$

PP-3



(a) Determine upper and lower threshold voltage

(b) Determine the period, time const, and the duty cycle of the square wave generator, given that the freq is 100Hz

(c) Draw the input-output voltage on the same graph

⇒ Soln:

$$(a) V_{UT} = \frac{R_2}{R_1 + R_2} V_H = 4V, V_{LT} = \frac{R_1}{R_1 + R_2} V_L = -2V$$

$$(b) f = 100\text{Hz}, T = \frac{1}{100} = 10\text{ms}$$

$$T_1 = R_f C_X \ln \left| \frac{V_H - V_{LT}}{V_H - V_{UT}} \right| = R_f C_X \ln 2$$

$$T_2 = R_f C_X \ln \left| \frac{V_L - V_{UT}}{V_L - V_{LT}} \right| = R_f C_X \ln 3$$

$$\Rightarrow T = T_1 + T_2 = R_f C_X (\ln 2 + \ln 3)$$

$$\therefore \tau = R_f C_X = \frac{T}{\ln 2 + \ln 3} = \frac{10\text{ms}}{\ln 2 + \ln 3} = 5.58\text{ms}$$

$$\therefore \text{Time const} \rightarrow \tau = 5.58\text{ms} \quad \therefore T_1 = \tau \ln 2 = 3.86$$

$$\therefore T_2 = 10 - 3.86 = 6.13$$

$$\text{Duty cycle, } D = \frac{T_1}{T} \times 100\%$$

$$D = \frac{R_f C_X \ln 2}{T} \times 100\%$$

$$= \frac{5.58 \times \ln 2}{10} \times 100\%$$

$$\therefore D = 38.67\%$$

